

-- HDOC DAY – 7 --

Question 1: Frustum of a Cone

1: Problem Understanding

- **Problem Statement:** Write a C program to calculate the Curved Surface Area (CSA), Total Surface Area (TSA), and Volume of a frustum of a cone given its top radius (r_1), bottom radius (r_2), and height (h).
- **Inputs:**
 - Top radius (r_1)
 - Bottom radius (r_2)
 - Height (h)
- **Outputs:**
 - Curved Surface Area (CSA)
 - Total Surface Area (TSA)
 - Volume (V)
- **Formulas:**
 - Slant height, $l = \sqrt{h^2 + (r_1 - r_2)^2}$
 - Curved surface area, $CSA = \pi * l * (r_1 + r_2)$
 - Total surface area, $TSA = CSA + \pi * r_1^2 + \pi * r_2^2$
 - Volume, $V = \frac{1}{3} * \pi * h * (r_1^2 + r_2^2 + r_1 * r_2)$

2: Test Case Table

Test case	input	Expected output	Actual output	Result (pass/fail)
1.	$r_1 = 3$	Curved Surface Area = 112.18	Curved Surface Area = 112.18	pass

	$r_2 = 5$ $h = 4$	Total Surface Area = 219.00 Volume = 205.25	Total Surface Area = 219.00 Volume = 205.25	
2.	$r_1 = 2$ $r_2 = 4$ $H = 3$	Curved Surface Area = 68.08 Total Surface Area = 130.91 Volume = 87.96	Curved Surface Area = 68.08 Total Surface Area = 130.91 Volume = 87.96	pass
3.	$r_1 = 6$ $r_2 = 10$ $h = 8$	Curved Surface Area = 448.71 Total Surface Area = 875.99 Volume = 1642.02	Curved Surface Area = 448.71 Total Surface Area = 875.99 Volume = 1642.02	pass

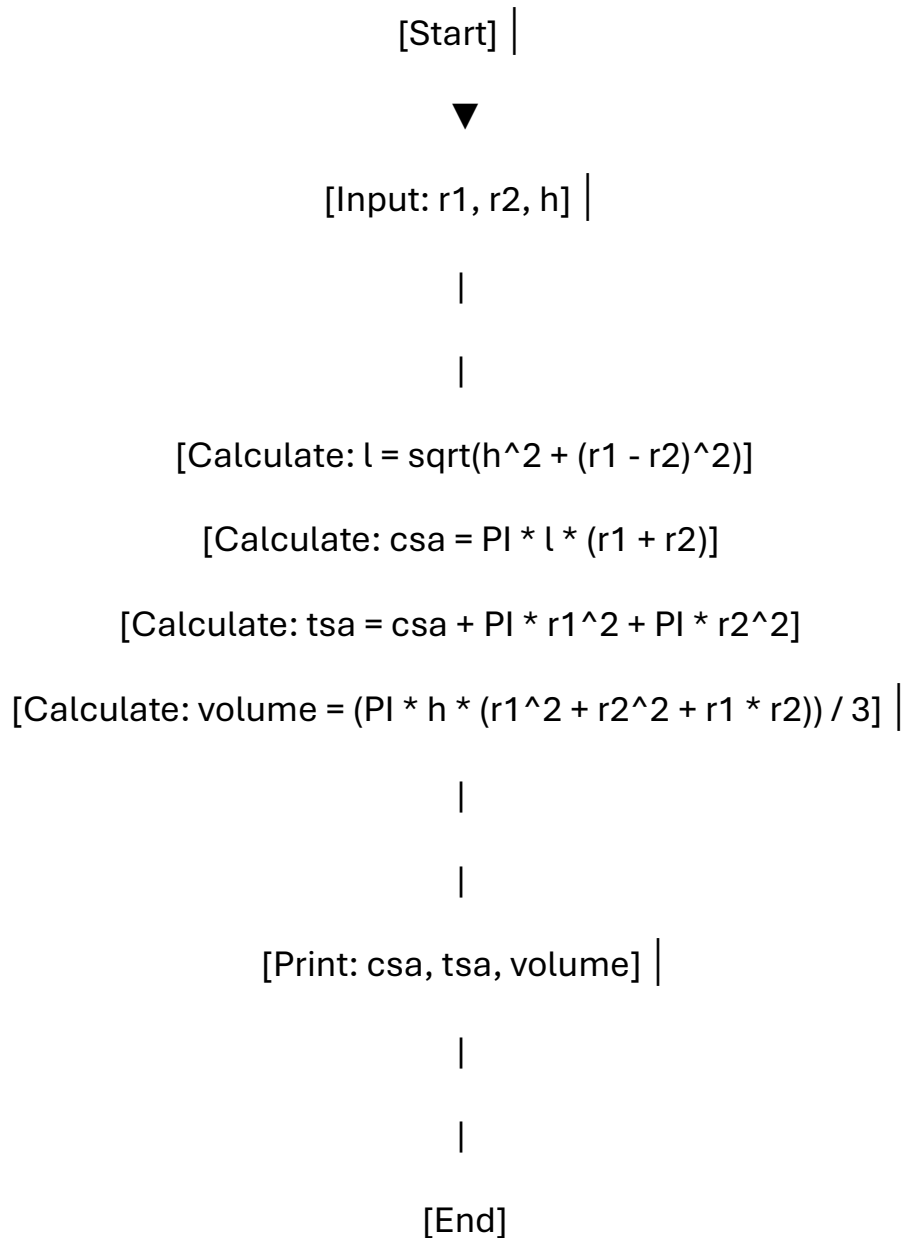
3: Algorithm :

- Start
- Declare variables ; r_1 , r_2 , h , slant_height , csa , tsa, volume
- Read values of r_1 , r_2 , and h from the user
- Calculate slant height , $l = \sqrt{h^2 + (r_1 - r_2)^2}$
- Calculate CSA : $CSA = \pi * l * (r_1 + r_2)$
- Calculate TSA : $TSA = CSA + \pi * r_1^2 + \pi * r_2^2$
- Calculate volume V : $V = \frac{1}{3} * \pi * h * (r_1^2 + r_2^2 + r_1 * r_2)$
- Print the calculated CSA, TSA, and volume.
- Stop.

ALGORITHM EVALUTION :

- Test case 1 verification :
 - $L = \sqrt{4^2 + (3-5)^2} = \sqrt{16 + 4} = \sqrt{20} = 4.472$
 - $CSA = \pi * 4.472 * (3+5) = 112.18$ (matches expected result)

4: Flowchart Representation



Flowchart Evaluation:

- **Test Case 1 Verification:** The flow follows sequential execution correctly. The calculations match step-by-step with expected mathematical formulas, successfully computing CSA=112.18, TSA=219.00, and Volume=205.25.

5: C Program Code

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main() {
```

```
    float r1, r2, h;
```

```
    float l, csa, tsa, vol;
```

```
    float pi = 3.14;
```

```
    // take input from user
```

```
    printf("Enter r1: ");
```

```
    scanf("%f", &r1);
```

```
    printf("Enter r2: ");
```

```
    scanf("%f", &r2);
```

```
    printf("Enter height: ");
```

```
    scanf("%f", &h);
```

```
    // calculations
```

```
    l = sqrt((h * h) + ((r1 - r2) * (r1 - r2)));
```

```
    csa = pi * l * (r1 + r2);
```

```

    tsa = csa + (pi * r1 * r1) + (pi * r2 * r2);
    vol = (pi * h * (r1 * r1 + r2 * r2 + r1 * r2)) / 3.0;

    // printing results
    printf("\nCurved Surface Area = %.2f\n", csa);
    printf("Total Surface Area = %.2f\n", tsa);
    printf("Volume = %.2f\n", vol);

    return 0;

}

```

6: Compilation, Execution

Execution Status: Compilation successful without any warnings or errors.

- **Evaluation Against Test Cases:**
 - Input: 3 5 4 → Outputs match expected values (112.18,219.00,205.25).
 - Input: 2 4 3 → Outputs match expected values (68.08,130.91,87.96).
 - Input: 6 10 8 → Outputs match expected values (448.71,875.99,1642.02).

7: Execution Screenshots

```
main.c
1 // ...
2
3 // ...
4
5 // ...
6 float l, csa, tsa, vol;
7 float pi = 3.14159;
8
9 // take input from user
10 printf("Enter r1: ");
11 scanf("%f", &r1);
12
13 printf("Enter r2: ");
14 scanf("%f", &r2);
15
16 printf("Enter height: ");
17 scanf("%f", &h);
18
19 // calculations
20 l = sqrt((h * h) + ((r1 - r2) * (r1 - r2)));
21
22 csa = pi * l * (r1 + r2);
23 tsa = csa + (pi * r1 * r1) + (pi * r2 * r2);
24 vol = (pi * h * (r1 * r1 + r2 * r2 + r1 * r2)) / 3.0;
25
26 // printing results
27 printf("\nCurved Surface Area = %.2f\n", csa);
28 printf("Total Surface Area = %.2f\n", tsa);
29 printf("Volume = %.2f\n", vol);
30
31 return 0;
32
```

Enter r2: 5
Enter height: 4

Curved Surface Area = 112.40
Total Surface Area = 219.21
Volume = 205.25

...Program finished with exit code 0
Press ENTER to exit console.

```
main.c
1 // ...
2
3 // ...
4
5 // ...
6 float l, csa, tsa, vol;
7 float pi = 3.14159;
8
9 // take input from user
10 printf("Enter r1: ");
11 scanf("%f", &r1);
12
13 printf("Enter r2: ");
14 scanf("%f", &r2);
15
16 printf("Enter height: ");
17 scanf("%f", &h);
18
19 // calculations
20 l = sqrt((h * h) + ((r1 - r2) * (r1 - r2)));
21
22 csa = pi * l * (r1 + r2);
23 tsa = csa + (pi * r1 * r1) + (pi * r2 * r2);
24 vol = (pi * h * (r1 * r1 + r2 * r2 + r1 * r2)) / 3.0;
25
26 // printing results
27 printf("\nCurved Surface Area = %.2f\n", csa);
28 printf("Total Surface Area = %.2f\n", tsa);
29 printf("Volume = %.2f\n", vol);
30
31 return 0;
32
```

Enter r1: 2
Enter r2: 4
Enter height: 3

Curved Surface Area = 67.96
Total Surface Area = 130.79
Volume = 87.96

...Program finished with exit code 0

```
main.c
6  float l, csa, tsa, vol;
7  float pi = 3.14159;
8
9  // take input from user
10 printf("Enter r1: ");
11 scanf("%f", &r1);
12
13 printf("Enter r2: ");
14 scanf("%f", &r2);
15
16 printf("Enter height: ");
17 scanf("%f", &h);
18
19 // calculations
20 l = sqrt((h * h) + ((r1 - r2) * (r1 - r2)));
21
22 csa = pi * l * (r1 + r2);
23 tsa = csa + (pi * r1 * r1) + (pi * r2 * r2);
24 vol = (pi * h * (r1 * r1 + r2 * r2 + r1 * r2)) / 3.0;
25
26 // printing results
27 printf("\nCurved Surface Area = %.2f\n", csa);
28 printf("Total Surface Area = %.2f\n", tsa);
29 printf("Volume = %.2f\n", vol);
30
31 return 0;
32
Enter r1: 6
Enter r2: 10
Enter height: 8
Curved Surface Area = 449.59
Total Surface Area = 876.84
Volume = 1642.00
...Program finished with exit code 0
```

Question 2: Area of a Quadrilateral

1: Problem Understanding

- **Problem Statement:** Write a C program to compute the total area of a quadrilateral defined by its four vertices (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , and (x_4, y_4) in order.
- **Inputs:**
 - (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , (x_4, y_4)
- **Outputs:**
 - Area of the Quadrilateral
- **Formula (Shoelace Formula):**
 - $$\text{area} = 0.5 \times (x_1 y_2 + x_2 y_3 + x_3 y_4 + x_4 y_1) - (y_1 x_2 + y_2 x_3 + y_3 x_4 + y_4 x_1)$$
 -

2: Test Case Table

Test case	Input(s)	Expected output(s)	Actual output(s)	Result (pass/fail)
1.	x1 = 0, y1 = 0 x2 = 4, y2 = 0 x3 = 4, y3 = 3 x4 = 0, y4 = 3	Area of Quadrilateral = 12.00	Area of Quadrilateral = 12.00	pass
2.	x1 = 1, y1 = 1 x2 = 5, y2 = 1 x3 = 4, y3 = 4 x4 = 0, y4 = 3	Area of Quadrilateral = 10.50	Area of Quadrilateral = 10.50	pass
3.	x1 = -2, y1 = 1 x2 = 2, y2 = -3 x3 = 5, y3 = 2 x4 = 0, y4 = 5	Area of Quadrilateral = 29.00	Area of Quadrilateral = 29.00	pass

3: Algorithm

- Start
- Declare variables for vertex coordinates: x1, y1, x2, y2, x3, y3, x4, y4 and double variable area.

- Read all 4 pairs of coordinates from the user.
- Calculate area using the Shoelace formula:

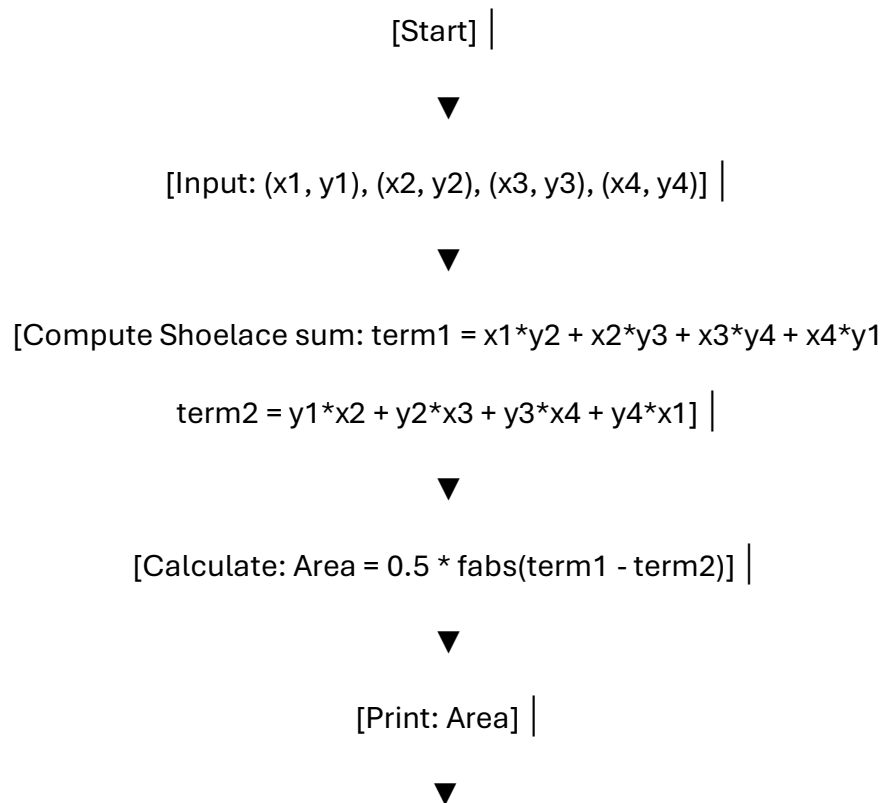
$$\text{area} = 0.5 \times |(x_1 y_2 + x_2 y_3 + x_3 y_4 + x_4 y_1) - (y_1 x_2 + y_2 x_3 + y_3 x_4 + y_4 x_1)|$$

- Print the computed area.
- Stop

ALGORITHM EVALUTION

- **Test Case 1 Verification:**
 - Vertices: (0,0),(4,0),(4,3),(0,3)
 - Part 1 = $(0 \cdot 0 + 4 \cdot 3 + 4 \cdot 3 + 0 \cdot 0) = 24$
 - Part 2 = $(0 \cdot 4 + 0 \cdot 4 + 3 \cdot 0 + 3 \cdot 0) = 0$
 - Area = $0.5 \times |24 - 0| = 12.00$ (Matches expected result)

4: Flowchart Representation



[End]

Flowchart evaluation :

Test case 1 : Evaluated logically step-by-step; correctly computes absolute area value resulting in 12.00 unit area.

5: C Program

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main() {
```

```
    float x1, y1, x2, y2, x3, y3, x4, y4;
```

```
    float sum1, sum2, area;
```

```
    // input vertex coordinates
```

```
    printf("Enter x1 and y1: ");
```

```
    scanf("%f %f", &x1, &y1);
```

```
    printf("Enter x2 and y2: ");
```

```
    scanf("%f %f", &x2, &y2);
```

```
    printf("Enter x3 and y3: ");
```

```
    scanf("%f %f", &x3, &y3);
```

```
    printf("Enter x4 and y4: ");
```

```
    scanf("%f %f", &x4, &y4);
```

```
    // shoelace formula calculation
```

```
    sum1 = (x1 * y2) + (x2 * y3) + (x3 * y4) + (x4 * y1);
```

```
    sum2 = (y1 * x2) + (y2 * x3) + (y3 * x4) + (y4 * x1);
```

```
    area = 0.5 * fabs(sum1 - sum2);
```

```

printf("\nArea of Quadrilateral = %.2f\n", area);

return 0;

}

```

6: Compilation, Execution

- **Execution Status:** Successfully compiled with 0 errors.
- **Evaluation Against Test Cases:**
 - Input: (0,0), (4,0), (4,3), (0,3) → Output: 12.00
 - Input: (1,1), (5,1), (4,4), (0,3) → Output: 10.50
 - Input: (-2,1), (2,-3), (5,2), (0,5) → Output: 29.00

7: Execution Screenshots

The screenshot displays a C program for calculating the area of a quadrilateral using the shoelace formula. The source code is shown in a dark-themed editor with line numbers 4 to 30. The program prompts the user to enter the four vertices (x1, y1), (x2, y2), (x3, y3), and (x4, y4). It then calculates the area using the formula: $area = 0.5 * \text{fabs}(sum1 - sum2)$, where $sum1 = (x1 * y2) + (x2 * y3) + (x3 * y4) + (x4 * y1)$ and $sum2 = (y1 * x2) + (y2 * x3) + (y3 * x4) + (y4 * x1)$. The output shows the area as 12.00 for the input (0,0), (4,0), (4,3), (0,3). The console window at the bottom shows the program's execution, including the prompts, user input, and the final output.

```

4 int main() {
5     float x1, y1, x2, y2, x3, y3, x4, y4;
6     float sum1, sum2, area;
7
8     // input vertex coordinates
9     printf("Enter x1 and y1: ");
10    scanf("%f %f", &x1, &y1);
11
12    printf("Enter x2 and y2: ");
13    scanf("%f %f", &x2, &y2);
14
15    printf("Enter x3 and y3: ");
16    scanf("%f %f", &x3, &y3);
17
18    printf("Enter x4 and y4: ");
19    scanf("%f %f", &x4, &y4);
20
21    // shoelace formula calculation
22    sum1 = (x1 * y2) + (x2 * y3) + (x3 * y4) + (x4 * y1);
23    sum2 = (y1 * x2) + (y2 * x3) + (y3 * x4) + (y4 * x1);
24
25    area = 0.5 * fabs(sum1 - sum2);
26
27    printf("\nArea of Quadrilateral = %.2f\n", area);
28
29    return 0;
30 }

```

Input

```

Enter x1 and y1: 0 0
Enter x2 and y2: 4 0
Enter x3 and y3: 4 3
Enter x4 and y4: 0 3

Area of Quadrilateral = 12.00

...Program finished with exit code 0
Press ENTER to exit console.

```

```
main.c
4 int main() {
5     float x1, y1, x2, y2, x3, y3, x4, y4;
6     float sum1, sum2, area;
7
8     // input vertex coordinates
9     printf("Enter x1 and y1: ");
10    scanf("%f %f", &x1, &y1);
11
12    printf("Enter x2 and y2: ");
13    scanf("%f %f", &x2, &y2);
14
15    printf("Enter x3 and y3: ");
16    scanf("%f %f", &x3, &y3);
17
18    printf("Enter x4 and y4: ");
19    scanf("%f %f", &x4, &y4);
20
21    // shoelace formula calculation
22    sum1 = (x1 * y2) + (x2 * y3) + (x3 * y4) + (x4 * y1);
23    sum2 = (y1 * x2) + (y2 * x3) + (y3 * x4) + (y4 * x1);
24
25    area = 0.5 * fabs(sum1 - sum2);
26
27    printf("\nArea of Quadrilateral = %.2f\n", area);
28
29    return 0;
30 }
```

Enter x1 and y1: 1 1
Enter x2 and y2: 5 1
Enter x3 and y3: 4 4
Enter x4 and y4: 0 3

Area of Quadrilateral = 10.50

...Program finished with exit code 0
Press ENTER to exit console.

```
main.c
4 int main() {
5     float x1, y1, x2, y2, x3, y3, x4, y4;
6     float sum1, sum2, area;
7
8     // input vertex coordinates
9     printf("Enter x1 and y1: ");
10    scanf("%f %f", &x1, &y1);
11
12    printf("Enter x2 and y2: ");
13    scanf("%f %f", &x2, &y2);
14
15    printf("Enter x3 and y3: ");
16    scanf("%f %f", &x3, &y3);
17
18    printf("Enter x4 and y4: ");
19    scanf("%f %f", &x4, &y4);
20
21    // shoelace formula calculation
22    sum1 = (x1 * y2) + (x2 * y3) + (x3 * y4) + (x4 * y1);
23    sum2 = (y1 * x2) + (y2 * x3) + (y3 * x4) + (y4 * x1);
24
25    area = 0.5 * fabs(sum1 - sum2);
26
27    printf("\nArea of Quadrilateral = %.2f\n", area);
28
29    return 0;
30 }
```

Enter x1 and y1: -2 1
Enter x2 and y2: 2 -3
Enter x3 and y3: 5 2
Enter x4 and y4: 0 5

Area of Quadrilateral = 29.00

...Program finished with exit code 0
Press ENTER to exit console.